

AYUSH SINGH

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🐙 GitHub

PROFESSIONAL SUMMARY

Machine Learning Engineer skilled in Machine Learning, Deep Learning, Natural Language Processing (NLP), Predictive Analytics, and MLOps with hands-on experience in building end-to-end AI and data-driven solutions. Proficient in Python, Scikit-learn, XGBoost, TensorFlow, MongoDB, MLflow, and Docker with expertise in data preprocessing, feature engineering, model training, optimization, and deployment.

TECHNICAL SKILLS

Programming Languages: Python, C++

Machine Learning & AI: Scikit-learn, XGBoost, TensorFlow, NLTK, Gensim

LLM & MLOps Tools: LangChain, LangSmith, FAISS, Groq API, MLflow, Docker

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Web Development & Frameworks: Flask, Streamlit, Beautiful Soup

Databases & Tools: MongoDB, MySQL, SQLite, Git

PROJECTS

AI-Powered Resume Optimizer | LangChain, FAISS, Groq API, Docker 🐙

- Developed an end-to-end **RAG-based resume optimization system** using **LangChain**, **FAISS**, and **LLaMA-3.1-8B-Instant** via **Groq API** to analyze resumes against target job descriptions
- Built intelligent **ATS score analysis**, keyword extraction, section-wise feedback, and AI-powered resume enhancement features to improve resume-job matching and recruiter visibility
- Integrated **LangSmith** for tracing and debugging, utilized **Nomic Embeddings** for semantic similarity search, and implemented **Docker**-based containerization for scalable deployment

AI Website Analyzer - Phishing Detection | Random Forest, MongoDB, MLflow, MLOps 🐙

- Developed an end-to-end **AI-powered phishing detection system** using **Random Forest Classification** achieving **97% prediction accuracy** through automated website feature analysis
- Designed a scalable **ETL and data preprocessing pipeline** integrating **MongoDB**, **KNN Imputer**, and **SMOTE** oversampling to handle missing values and class imbalance efficiently
- Implemented an end-to-end **MLOps pipeline** using **MLflow**, automated model training, **Beautiful Soup**-based web scraping, and **data drift detection reporting** for continuous monitoring

AI Text Detector | NLP, Word2Vec, Gensim 🐙

- Developed an end-to-end **NLP-based AI text detection system** using custom-trained **Word2Vec** embeddings with **Gensim** for semantic feature extraction and intelligent classification
- Built an advanced **text preprocessing and feature engineering pipeline** including tokenization, stemming, normalization, and linguistic processing using **Snowball Stemmer**
- Trained and optimized a high-performance **Decision Tree Classification model** achieving **97.7% accuracy** for AI-generated text detection and automated content verification

EDUCATION

VIT Bhopal University

B.Tech CSE with Specialisation in AIML - CGPA: 9.05

2023 – 2027

Bhopal, India

Gyan Devi Senior Secondary School

CBSE 12th Board - 94.6%

2023

Gurgaon, India

ACHIEVEMENTS

- Shell Data Science Hackathon:** Secured **Top 200** position among **1400+** teams
- Zelestra Data Science Hackathon:** Achieved **Top 200** rank among **1200+** participating teams
- Data Structures & Algorithms:** Solved **400+** coding problems across **LeetCode** and **GeeksforGeeks**
- School Head Boy:** Selected as **Head Boy** at Gyan Devi Senior Secondary School

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- NPTEL Cloud Computing
- HackerRank Python (Basic)